RAJALAKSHMI ENGINEERING COLLEGE
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CS23431 - OPERATING SYSTEM

EXP 6: FIFO Page Replacement Algorithm Using C

QUESTION :

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits. .

CODE:

```c
#include <stdio.h>

int main() {
    int n, frames;

    printf("Enter number of pages in reference string: \n");
    scanf("%d", &n);

    int ref[n];
    printf("Enter the reference string: \n");
    for(int i = 0; i < n; i++) {
        scanf("%d", &ref[i]);
    }

    printf("Enter number of frames: \n");
    scanf("%d", &frames);

    int frame[frames];
    int front = 0;
    int faults = 0, hits = 0;

    // Initialize frames with -1
    for(int i = 0; i < frames; i++)
        frame[i] = -1;

    for(int i = 0; i < n; i++) {
        int found = 0;

        // Check for hit
        for(int j = 0; j < frames; j++) {
            if(frame[j] == ref[i]) {
```

```
                    found = 1;
                    hits++;
                    break;
                }
            }
    }
```

Output:

Custom Test Case

Custom Test — Success

Input :

```
7
1 3 0 3 5 6 3
3
```

Expected Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1
```

Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1
```